

Supplementary Material

CXT-Fish: Group-Disjoint Evaluation and Foreground-Sufficiency Training for Context-Robust Underwater Fish Recognition

All supplementary analyses are based on frozen predictions, manifests, and metadata. No model was trained or re-inferred for this package; official Fish4Knowledge TEST was not accessed.

S1. Group-weighted robustness sensitivity

The manuscript's primary cross-class composite macro-F1 is class-balanced but image-weighted within species. This sensitivity gives each recorded group equal weight within species and then gives species equal weight. The nine fold–seed cells are averaged with equal weight. It is descriptive and does not replace the primary macro-F1 estimand.

Metric	F0 mean $\pm$ SD	CXT-Fish mean $\pm$ SD	Δ (pp)	Favourable cells
group balanced accuracy	0.6979 $\pm$ 0.0278	0.7426 $\pm$ 0.0161	+4.47	8/9
species group balanced accuracy	0.5271 $\pm$ 0.0156	0.5944 $\pm$ 0.0269	+6.73	9/9

Table S1. Cross-class donor-context-composite group-weighted sensitivity. SD is the sample standard deviation across nine equally weighted fold–seed cells. Official-test access: false.

S2. Recorded-group counts by species and outer fold

Species	Fold 1	Fold 2	Fold 3
1	1414	1413	1413
10	17	18	18
11	14	14	14
12	10	9	9
13	9	9	9
14	7	8	8
15	6	5	5
16	5	5	5
2	408	409	408
3	392	391	392
4	340	341	340
5	179	178	179
6	26	27	26
7	24	23	24
8	23	24	24
9	20	19	19

Table S2. Recorded-group counts in the three frozen outer-test manifests. There are 48 fold $\times$ species strata; the minimum is 5 and the maximum is 1414. These finite clusters motivate cautious interpretation of percentile cluster intervals for sparsely represented strata.

Outer fold	Head species	Mid species	Tail species
1	1, 2, 3, 4, 5, 7	6, 8, 9, 10, 13	11, 12, 14, 15, 16
2	1, 2, 3, 4, 5, 7	6, 8, 9, 10, 13	11, 12, 14, 15, 16
3	1, 2, 3, 4, 5, 10	7, 8, 9, 13, 16	6, 11, 12, 14, 15

Table S2b. Fold-local class-frequency tiers used by the outer evaluators. Each mapping is derived deterministically from that fold's outer-training image frequencies; the tier labels are not imported from a single development split.

S3. Donor-species audit

Donor species	Recipients	Fraction	Donor groups	Correctness effect
1.0	22644	0.278	2716	+4.40 pp
10.0	1251	0.015	49	+10.47 pp
11.0	417	0.005	31	+4.80 pp
12.0	624	0.008	22	+1.60 pp
13.0	810	0.010	27	-2.10 pp
14.0	303	0.004	22	+1.98 pp
15.0	183	0.002	16	+0.00 pp
16.0	846	0.010	15	+10.28 pp
2.0	10890	0.134	1022	+4.55 pp
3.0	13899	0.171	1021	+6.33 pp
4.0	15225	0.187	880	+7.28 pp
5.0	9561	0.117	473	+0.33 pp
6.0	846	0.010	59	+3.31 pp
7.0	1860	0.023	64	+3.01 pp
8.0	951	0.012	63	+2.94 pp
9.0	1089	0.013	51	+0.18 pp

Table S3. Frozen seed-3407 donor-species audit. The donor-species-equal-weight correctness sensitivity is +3.71 percentage points; this is descriptive and not a replacement for the natural-manifest-weighted main result.

S4. MobileNetV3-Large fold-level sensitivity

Fold	Clean Δ pp	Foreground Δ pp	Cross composite Δ pp	DAR-flip Δ pp
1	-7.32	+4.69	-2.11	-5.45
2	-0.25	+12.99	+7.55	-2.32
3	-1.26	+12.99	+9.03	-6.41

Table S4. MobileNetV3-Large architecture-sensitivity fold effects at the pre-specified seed 3407. Foreground improves in 3/3 folds, cross-class composite improves in 2/3 folds, and DAR-flip decreases in 3/3 folds; clean effects are heterogeneous and negative on average.

S5. Evidence-role and reproducibility notes

- Main ResNet18 evidence: same-corpus frozen group-disjoint re-evaluation, 3 folds $\times$ 3 seeds (17, 2026, 3407).
- Primary estimand: equal-weight mean of nine unrounded fold-seed cross-class macro-F1 differences.
- Primary interval: 5,000 paired species-stratified group-cluster bootstrap replicates, conditional on completed development, frozen models/seeds, and frozen seed-3407 donor realization.
- Donor-realization, F0-2RGB, donor-subject-suppressed, group-weighted, and MobileNet analyses: post-hoc validity or boundary evidence; not independent validation.
- Route C: fixed mechanism stress control, not faithful CLIB reproduction and not an exhaustive contrastive-learning comparison; detailed values are in Supplementary Note S6.
- Higher acquisition units (camera/session/deployment/date) were not verifiable from frozen recognition metadata.
- Official Fish4Knowledge TEST accessed: false.
- The repository retains stopped TCCR, DTH, donor-aware, TRAFS, CIR, DLE, TAP, and CXT-Select explorations as negative-result records.

S6. Route C mechanism stress control

Supplementary Note S6. Route C is a separately frozen two-stage mask-guided subject-non-primary contrastive mechanism control. It completed 9/9 fold-seed cells with official-test access false in every metrics file. Its equal-weight means were 0.6984 original-view macro-F1, 0.6981 foreground macro-F1, 0.7014 same-class-composite macro-F1, 0.4224 cross-class-composite macro-F1, and 0.1610 DAR-flip. Route C is retained as a fixed stress control and is not interpreted as a faithful CLIB reproduction or an exhaustive contrastive-learning comparison.

S7. Donor residual-bin heterogeneity

Residual bin	Pairs	Recipients	F0 correctness	CXT-Fish correctness	Delta (pp)	F0 DAR-flip	CXT DAR-flip
(0,1%]	5652	1884	0.849	0.882	+3.27 pp	0.002	0.002
(1%,5%]	67083	22361	0.726	0.771	+4.52 pp	0.001	0.001
(5%,10%]	8586	2862	0.520	0.595	+7.51 pp	0.001	0.001
>10%	78	26	0.372	0.423	+5.13 pp	0.000	0.000

Table S7. Frozen donor-subject residual-bin sensitivity. Bins are defined by donor foreground fraction outside the recipient hard mask; empty bins are not shown. Values are descriptive and do not define a new inferential endpoint.

S8. Donor-species equal-weight sensitivity

Donor species	Recipients	Donor groups	F0 correctness	CXT-Fish correctness	Delta (pp)
1.0	22644	2716	0.725	0.769	+4.40 pp
10.0	1251	49	0.748	0.853	+10.47 pp
11.0	417	31	0.863	0.911	+4.80 pp
12.0	624	22	0.870	0.886	+1.60 pp
13.0	810	27	0.681	0.660	-2.10 pp
14.0	303	22	0.904	0.924	+1.98 pp
15.0	183	16	0.962	0.962	+0.00 pp
16.0	846	15	0.796	0.898	+10.28 pp
2.0	10890	1022	0.832	0.878	+4.55 pp
3.0	13899	1021	0.779	0.843	+6.33 pp
4.0	15225	880	0.495	0.568	+7.28 pp
5.0	9561	473	0.672	0.676	+0.33 pp
6.0	846	59	0.888	0.921	+3.31 pp
7.0	1860	64	0.881	0.911	+3.01 pp
8.0	951	63	0.877	0.906	+2.94 pp
9.0	1089	51	0.903	0.904	+0.18 pp

Table S8. Donor-species equal-weight correctness sensitivity. Species are weighted equally for this descriptive audit; the natural-manifest-weighted main result remains primary.